Beibin Li

Curriculum Vitae

(Updated Jul., 2026)
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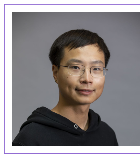

Full-time Experience

2026 – Now **Lead Scientist**, *Apodex*

Frontier AI lab building general-purpose reasoning systems and research agents. Leading the **Coding** team and the **Self-Evolve** team.

2025 – 2026 **Member of Technical Staff**, *xAI*

Reasoning team. Post-training and reinforcement learning for the Grok frontier model family, with a focus on reasoning, agents, and tool use.

Lead	Grok Code 4.3 and Grok Build (Models)
Co-Owner	Grok 4.1 Fast, Grok 4.2 API model (**1T+ tokens** per month); led Tool Call and enterprise model training
Contributor	Grok 4.1, Grok Code/Build, Grok in Tesla, Grok for SpaceX

2024 – 2025 **Researcher**, *Meta*

Meta GenAI, Llama team. Post-trained LLMs with reinforcement learning for reasoning, safety, and user experience.

Generative Judge	Lead	Designed generative judges with rubrics to improve reasoning, instruction-following, and tool use while reducing factual errors during online RL fine-tuning.
Human Alignment	Lead	Built an automated data-flywheel for human preference alignment, integrating synthetic data generation and continuous RM/LLM retraining to drive DAU growth.
Multilingual	Engineer	Scaled eval 10× across 55 languages with synthetic datasets and LLM judges for cross-lingual coverage.
Inference System	Researcher	Co-designed dynamic routing with speculative decoding, cutting latency by 30%+ and boosting token throughput.

2022 – 2024 **Senior Research Engineer**, *Microsoft Research*

Machine learning and optimization (MLO) group. Leading LLM, DL, and optimization research for language models, supply chain, and cloud system.

🤖 AutoGen	Co-Author	Earned **30,000+ GitHub stars** for *Next-Gen LLM Apps via Multi-Agent Conversation* by building core agents (code, multimodal, Gemini) and driving the development of reflection, evaluation, memory, RAG, and AG Studio.
📦 OptiGuide	First Author	Developed *LLM for Combinatorial Optimization*, architecting the end-to-end research and engineering pipeline and reducing manual engineering effort by over 40%.
🏆 MLSys Best Paper	First Author	Received the **Best Paper Award** at MLSys for *VM Allocation with Lifetime Prediction*, implementing the model and evaluation framework in PyTorch and C++.
♾ MILP-Evolve	Lead Author	Introduced *Foundation Models for Mixed Integer Linear Programming*, designing data processing and training workflows in PyTorch, C++, and MILP solvers.
🧠 Reflect-RL	Lead Author	Developed *Two-Player Online RL Fine-Tuning for Language Models*, pioneering agentic RL for LLMs using PyTorch and Gym.

2015 – 2017 **Researcher**, *Seattle Children's Research Institute; Yale University*
Advisor: Frederick Shic
Developed deep learning systems specialized for image and video analysis. Eye-tracking research, including fixation identification algorithms, spatio-temporal analysis, etc.
📱 Social Influences on EF in Autism: Design of a Mobile Gaming Platform
👓 Eye-Tracking: the Autism Biomarkers Consortium for Clinical Trials

Education

2022 **Ph.D. in Computer Science and Engineering**, *University of Washington, Seattle*, *Faithful Steward Endowed Fellowship*; Advisors: Linda Shapiro and Frederick Shic
🎓 *Low-Resource Neural Adaptation: A Unified Data Adaptation Framework for Neural Networks*

2015 **Bachelor of Science (Dual Major), Computer Science and Mathematics**, *University of Michigan, Ann Arbor*, Graduated with *University Honors*

Publications

Bae, Y., **Li, B**, Dommer, K. J., Reninger, M., Hegerberg, K., Ajwani, A., et al. (2025). Genetic associations of blink behavior in infants and toddlers: A computer vision approach. *ACM Symposium on Eye Tracking Research and Applications (ETRA).*

Bakhtiarvand, T., Kalita, J., Foster, C., Barney, E., Kim, M., **Li, B**, Ventola, P., et al. (2025). Exploring gaze patterns for autism diagnosis with lstm and attention mechanisms. *International Society for Autism Research (INSAR) Annual Meeting.*

Ghezloo, F., Seyfioglu, M. S., Soraki, R., Ikezogwo, W. O., **Li, B**, Vivekanandan, T., Elmore, J. G., Krishna, R., & Shapiro, L. (2025). Pathfinder: A multi-modal multi-agent system for medical diagnostic decision-making applied to histopathology. *IEEE/CVF International Conference on Computer Vision (ICCV).*

Hegerberg, K. E., Oien, R. A., Dommer, K. J., Skytta, J., Reninger, M. C., Ajwani, A., **Li, B**, et al. (2025). Non-social visual search advantage in asd emerges in toddlerhood: A longitudinal study. *International Society for Autism Research (INSAR) Annual Meeting*.

Huang, Y., Liu, Y., Gunawi, H. S., **Li, B**, & Hwang, C. (2025). Alchemist: Towards the design of efficient online continual learning system. *arXiv preprint arXiv:2503.01066*.

Li, S., Kulkarni, J., Menache, I., Wu, C., & **Li, B**. (2025). Towards foundation models for mixed integer linear programming. *International Conference on Learning Representations (ICLR)*.

Liu, R. P., Mellou, K., Gong, X.-Y., **Li, B**, Coffee, T., Pathuri, J., Simchi-Levi, D., & Menache, I. (2025). Efficient cloud server deployment under demand uncertainty. *Manufacturing & Service Operations Management*.

Llama Team, Meta AI. (2025). *The llama 4 herd: The beginning of a new era of natively multimodal ai innovation* (tech. rep.). Meta.

Wall, C. A., Hudac, C., Dommer, K. J., **Li, B**, Atyabi, A., Foster, C., Wang, Q., Barney, E., et al. (2025). Preserved but un-sustained responses to bids for dyadic engagement in school-age children with autism. *Journal of Autism and Developmental Disorders*, 1–9.

Ye, G., Pham, K. D., Zhang, X., Gopi, S., Peng, B., **Li, B**, Kulkarni, J., & Inan, H. A. (2025). On the emergence of thinking in llms i: Searching for the right intuition. *arXiv preprint arXiv:2502.06773*.

Zhang, S., Yin, M., Zhang, J., Liu, J., Han, Z., Zhang, J., **Li, B**, Wang, C., Wang, H., Chen, Y., & Wu, Q. (2025). Which agent causes task failures and when? on automated failure attribution of llm multi-agent systems [Spotlight paper (top 2.6%)]. *International Conference on Machine Learning (ICML)*.

Karimi, P., Pirelli, S., Kakarla, S. K. R., Beckett, R., Segarra, S., **Li, B**, Namyar, P., & Arzani, B. (2024). Towards safer heuristics with xplain. *ACM Workshop on Hot Topics in Networks (HotNets)*.

Lu, Y., Bian, S., Chen, L., He, Y., Hui, Y., Lentz, M., **Li, B**, Liu, F., Li, J., Liu, Q., Liu, R., et al. (2024). Computing in the era of large generative models: From cloud-native to ai-native. *arXiv preprint arXiv:2401.12230*.

Li, B, Zhang, Y., Bubeck, S., Pathuri, J., & Menache, I. (2024). Small language models for application interactions: A case study. *arXiv preprint arXiv:2405.20347*.

Wu, Q., Bansal, G., Zhang, J., Wu, Y., Zhang, S., Zhu, E., **Li, B**, Jiang, L., Zhang, X., & Wang, C. (2024). Autogen: Enabling next-gen llm applications via multi-agent conversation framework. *Conference on Language Modeling (COLM)*.

Zhou, R., Du, S. S., & **Li, B**. (2024). Reflect-RL: Two-Player Online RL Fine-Tuning for LMs. *Annual Meeting of the Association for Computational Linguistics (ACL)*.

Atyabi, A., Boccanfuso, L., Snider, J., Kim, M., Barney, E., Ahn, Y. A., **Li, B**, Dommer, K. J., & Shic, F. (2023). Large-scale investigations of aac usage patterns: Trends, autism, and stacked autoencoders. *2023 IEEE 13th Annual Computing and Communication Workshop and Conference (CCWC)*, 851–859.

Atyabi, A., Shic, F., Jiang, J., Foster, C. E., Barney, E., Kim, M., **Li, B**, Ventola, P., & Chen, C. H. (2023). Stratification of children with autism spectrum disorder through fusion of temporal information in eye-gaze scan-paths. *ACM Transactions on Knowledge Discovery from Data*, *17*(2), 1–20.

Dong, M., Telesca, D., Sugar, C., Shic, F., Naples, A., Johnson, S. P., **Li, B**, Atyabi, A., Xie, M., Webb, S. J., et al. (2023). A functional model for studying common trends across trial time in eye tracking experiments. *Statistics in biosciences*, *15*(1), 261–287.

Hochhauser, M., Dommer, K. J., Atyabi, A., **Li, B**, Ahn, Y. A., Aubertine, M., Kim, M., Corrigan, S. G., Pelphrey, K. A., & Shic, F. (2023). Comparing attention to biological motion in autism across age groups using eye-tracking. *Proceedings of the 2023 Symposium on Eye Tracking Research and Applications*, 1–3.

Liu, K., **Li, B**, Wu, W., May, C., Chang, O., Knezevich, S., Reisch, L., Elmore, J., & Shapiro, L. (2023). Vsgd-net: Virtual staining guided melanocyte detection on histopathological images. *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 1918–1927.

Sajal, S. M., Marshall, L., **Li, B**, Zhou, S., Pan, A., Mellou, K., Narayanan, D., Zhu, T., Dion, D., Moscibroda, T., et al. (2023). Kerveros: Efficient and scalable cloud admission control. *17th USENIX Symposium on Operating Systems Design and Implementation (OSDI 23)*, 227–245.

Shic, F., Barney, E. C., Naples, A. J., Dommer, K. J., Chang, S. A., **Li, B**, McAllister, T., Atyabi, A., Wang, Q., Bernier, R., et al. (2023). The selective social attention task in children with autism spectrum disorder: Results from the autism biomarkers consortium for clinical trials (abc-ct) feasibility study. *Autism Research*, *16*(11), 2150–2159.

Shic, F., Dommer, K., Benton, J., **Li, B**, Snider, J. C., Nyström, P., & Falck-Ytter, T. (2023). Remote, tablet-based assessment of gaze following: A nationwide infant twin study.

Li, B, Barbalho, H., Kovaleski, P., Marshall, L., Molinaro, M., Pan, A., Cortez, E., Leao, M., Patwari, H., Tang, Z., et al. (2023). Virtual machine allocation with lifetime predictions. *Proceedings of Machine Learning and Systems*, *5*.

Li, B, Mellou, K., Zhang, B., Pathuri, J., & Menache, I. (2023). Large language models for supply chain optimization. *arXiv preprint arXiv:2307.03875*.

Xiao, L., Zhu, G., Huang, K., Wen, S., Feng, L., **Li, B**, Xiao, B., Liu, D., & Wang, Q. (2023). Memory characteristics in mesial temporal lobe epilepsy: Insights from an eye tracking memory game and neuropsychological assessments. *CNS Neuroscience & Therapeutics*.

Marathe, K., Marasinou, C., **Li, B**, Nakhaei, N., Li, B., Elmore, J. G., Shapiro, L., & Hsu, W. (2022). Automated quantitative assessment of amorphous calcifications: Towards improved malignancy risk stratification. *Computers in biology and medicine*, *146*, 105504.

Nofallah, S., **Li, B**, Mokhtari, M., Wu, W., Knezevich, S., May, C. J., Chang, O. H., Elmore, J. G., & Shapiro, L. G. (2022). Improving the diagnosis of skin biopsies using tissue segmentation. *Diagnostics*, *12*(7), 1713.

Nuechterlein, N., **Li, B**, Shapiro, L., Cimino, P., & Holland, E. (2022). Genome-wide somatic copy number alteration signatures have prognostic implications for adult astrocytic glioma. *NEURO-ONCOLOGY*, *24*, 9–10.

Shic, F., Naples, A. J., Barney, E. C., Chang, S. A., **Li, B**, McAllister, T., Kim, M., Dommer, K. J., Hasselmo, S., Atyabi, A., et al. (2022). The autism biomarkers consortium for clinical trials:

Evaluation of a battery of candidate eye-tracking biomarkers for use in autism clinical trials. *Molecular Autism*, *13*(1), 1–17.

Li, B, Lu, Y., & Kandula, S. (2022). Warper: Efficiently adapting learned cardinality estimators to data and workload drifts. *Proceedings of the 2022 International Conference on Management of Data*, 1920–1933.

Li, B, Snider, J. C., Wang, Q., Mehta, S., Foster, C., Barney, E., Shapiro, L., Ventola, P., & Shic, F. (2022). Calibration error prediction: Ensuring high-quality mobile eye-tracking. *2022 Symposium on Eye Tracking Research and Applications*, 1–7.

Liu, K., Mokhtari, M., **Li, B**, Nofallah, S., May, C., Chang, O., Knezevich, S., Elmore, J., & Shapiro, L. (2021). Learning melanocytic proliferation segmentation in histopathology images from imperfect annotations. *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 3766–3775.

Mehta, S., Nuechterlein, N., Mercan, E., **Li, B**, Nofallah, S., Wu, W., Lu, X., Caspi, A., Rastegari, M., Elmore, J., et al. (2021). Applications of the espnet architecture in medical imaging. In *State of the art in neural networks and their applications* (pp. 117–131). Elsevier.

Nuechterlein, N., **Li, B**, Feroze, A., Holland, E. C., Shapiro, L., Haynor, D., Fink, J., & Cimino, P. J. (2021). Radiogenomic modeling predicts survival-associated prognostic groups in glioblastoma. *Neuro-oncology Advances*, *3*(1), vdab004.

Nuechterlein, N., **Li, B**, Seyfioğlu, M. S., Mehta, S., Cimino, P. J., & Shapiro, L. (2021). Leveraging unlabeled data for glioma molecular subtype and survival prediction. *2020 25th International Conference on Pattern Recognition (ICPR)*, 7149–7156.

Li, B, Lu, Y., Wang, C., & Kandula, S. (2021a). Cardinality estimation: Is machine learning a silver bullet. *3rd International Workshop on Applied AI for Database Systems and Applications (AIDB).*

Li, B, Lu, Y., Wang, C., & Kandula, S. (2021b). Q-error bounds of random uniform sampling for cardinality estimation. *arXiv preprint arXiv:2108.02715.*

Li, B, Mercan, E., Mehta, S., Knezevich, S., Arnold, C. W., Weaver, D. L., Elmore, J. G., & Shapiro, L. G. (2021). Classifying breast histopathology images with a ductal instance-oriented pipeline. *2020 25th International Conference on Pattern Recognition (ICPR)*, 8727–8734.

Li, B, Nuechterlein, N., Barney, E., Foster, C., Kim, M., Mahony, M., Atyabi, A., Feng, L., Wang, Q., Ventola, P., et al. (2021). Learning oculomotor behaviors from scanpath. *Proceedings of the 2021 International Conference on Multimodal Interaction*, 407–415.

Wang, H., **Li, B**, Shic, F., Hu, B., & Wang, Q. (2021). Comparing robustness and efficiency of closed eye detection in images. *2021 6th International Conference on Image, Vision and Computing (ICIVC)*, 6–12.

Zhu, G., Wang, J., Xiao, L., Yang, K., Huang, K., **Li, B**, Huang, S., Hu, B., Xiao, B., Liu, D., et al. (2021). Memory deficit in patients with temporal lobe epilepsy: Evidence from eye tracking technology. *Frontiers in Neuroscience*, *15*, 716476.

Jas, M., Achakulvisut, T., Idrizović, A., Acuna, D., Antalek, M., Marques, V., Odland, T., Garg, R. P., Agrawal, M., Umegaki, Y., et al. (2020). Pyglmnet: Python implementation of elastic-net regularized generalized linear models. *Journal of Open Source Software*, *5*(47).

Nuechterlein, N., **Li, B**, Fink, J., Haynor, D., Holland, E., Shapiro, L., & Cimino, P. (2020). Nimg-46. radiogenomic features predict clinically relevant genome-wide alteration signatures in glioblastoma. *Neuro-Oncology*, *22*(Suppl 2), ii158.

Li, B, Barney, E., Hudac, C., Nuechterlein, N., Ventola, P., Shapiro, L., & Shic, F. (2020). Selection of eye-tracking stimuli for prediction by sparsely grouped input variables for neural networks: Towards biomarker refinement for autism. *ACM Symposium on Eye Tracking Research and Applications*, 1–8.

Webb, S. J., Shic, F., Murias, M., Sugar, C. A., Naples, A. J., Barney, E., Borland, H., Hellemann, G., Johnson, S., Kim, M., et al. (2020). Biomarker acquisition and quality control for multi-site studies: The autism biomarkers consortium for clinical trials. *Frontiers in Integrative Neuroscience*, *13*, 71.

Wu, W., **Li, B**, Mercan, E., Mehta, S., Bartlett, J., Weaver, D. L., Elmore, J. G., & Shapiro, L. G. (2020). Mlcd: A unified software package for cancer diagnosis. *JCO clinical cancer informatics*, *4*, 290–298.

Li, B, Mehta, S., Aneja, D., Foster, C., Ventola, P., Shic, F., & Shapiro, L. (2019). A facial affect analysis system for autism spectrum disorder. *2019 IEEE International Conference on Image Processing (ICIP)*, 4549–4553.

Li, B, Nuechterlein, N., Barney, E., Hudac, C., Ventola, P., Shapiro, L., & Shic, F. (2019). Sparsely grouped input variables for neural networks. *arXiv preprint arXiv:1911.13068*.

Li, B, Atyabi, A., Kim, M., Barney, E., Ahn, A. Y., Luo, Y., Aubertine, M., Corrigan, S., St. John, T., Wang, Q., et al. (2018). Social influences on executive functioning in autism: Design of a mobile gaming platform. *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*, 1–13.

Atyabi, A., **Li, B**, Ahn, Y. A., Kim, M., Barney, E., & Shic, F. (2017). An exploratory analysis targeting diagnostic classification of aac app usage patterns. *2017 International Joint Conference on Neural Networks (IJCNN)*, 1633–1640.

Boccanfuso, L., Wang, Q., Leite, I., **Li, B**, Torres, C., Chen, L., Salomons, N., Foster, C., Barney, E., Ahn, Y. A., et al. (2016). A thermal emotion classifier for improved human-robot interaction. *2016 25th IEEE International Symposium on Robot and Human Interactive Communication (RO-MAN)*, 718–723.

Li, B, Boccanfuso, L., Wang, Q., Barney, E., Ahn, Y. A., Foster, C., Chawarska, K., Scassellati, B., & Shic, F. (2016). Human robot activity classification based on accelerometer and gyroscope. *2016 25th IEEE international symposium on robot and human interactive communication (RO-MAN). Presented at the 2016 25th IEEE international symposium on robot and human interactive communication (RO-MAN)*, 423–424.

Li, B, Wang, Q., Barney, E., Hart, L., Wall, C., Chawarska, K., de Urabain, I. S., Smith, T. J., & Shic, F. (2016). Modified dbscan algorithm on oculomotor fixation identification. *Proceedings of the Ninth Biennial ACM Symposium on Eye Tracking Research & Applications*, 337–338.

Li, B, Wang, Q., Boccanfuso, L., & Shic, F. (2016). Optimality of the distance dispersion fixation identification algorithm. *Proceedings of the Ninth Biennial ACM Symposium on Eye Tracking Research & Applications*, 339–340.

Wang, Q., Barney, E., Wall, C., DiNicola, L., Foster, C., Ahn, Y., **Li, B**, Katarzyna, C., & Shic, F. (2016). Hybrid calibration for eye tracking: Smooth pursuit trajectory with anchor points. *Journal of Vision*, *16*(12), 1355–1355.

Wang, Q., Boccanfuso, L., **Li, B**, Ahn, A. Y.-j., Foster, C. E., Orr, M. P., Scassellati, B., & Shic, F. (2016). Thermographic eye tracking. *Proceedings of the Ninth Biennial ACM Symposium on Eye Tracking Research & Applications*, 307–310.

Li, B, Boccanfuso, L., Valencia, S., Scassellati, B., & Shic, F. (2015). Background music and sound effects in human-robot interaction.

Awards

2017-2018	Faithful Steward Endowed Fellowship, University of Washington
2015-2017	Translational Technologies Fellowship, Yale University
2013–2015	University Honors, University of Michigan
2014	The Mathematical Contest in Modeling (MCM), Honorable Mention
2010-2013	Presidential Scholarship, Rhodes College

Academic Service

2022, 2023	**Committee Member**. ACM Symposium on Eye Tracking Research & Applications
2021, 2022	**Reviewer**. Computers in Biology and Medicine, Journal
2021	**Reviewer**. Mathematical Biosciences and Engineering, Journal
2021	**Reviewer**. Machine Learning in Computational Biology (NeurIPS)
2020	**Reviewer**. ACM European Conference on Information Systems (ECIS)
2020	**Reviewer**. International Conference on Mobile, Hybrid, and On-line Learning
2019, 2020, 2023	**Reviewer**. ACM SIGCHI Conference on Human Factors in Computing Systems
2018, 2020	**Reviewer**. ACM Symposium on Eye Tracking Research & Applications

Teaching

2021	CSE 577, Advanced Computer Vision: Medical Imaging.
2020	CSE 599, AI and the Brain.
2020	CSE 455, Computer Vision.
2019	CSE 473, Intro to Artificial Intelligence.
2018	CSE 546, Machine Learning.
2015	EECS 376, Theory of Computation